Granular Instrumental Variables

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The Law of Large Numbers fails in modern economies

In a standard economy, micro-shocks cancel out at a rate of $1/\sqrt{N}$ Proof.

In a **granular** economy, where the size distribution of firms follows a power law, a few firms are so large that their idiosyncratic luck drives aggregate volatility.

Endogenous aggregate shocks confound the effect of large firms

The Goal: Estimate the causal elasticity of wages to labor demand (ψ).

The Identification Challenge: **Simultaneity**

- **Supply Equation:** Aggregate wages respond to labor demand and macro supply shocks (ε_t).
- **Demand Equation:** Labor demand depends on wages and macro demand shocks (η_t).

The Problem	$\text{Cov}(L_{St}, \varepsilon_t) \neq 0$
The Reason	Common macro shocks (η_t, ε_t) drive both variables simultaneously, making OLS estimates of ψ biased and inconsistent.

The Simple Model

Difference-in-averages as an instrument

A simple model of aggregate and idiosyncratic shocks

Consider an economy with N firms and a single common factor (η_t).

The system is given by:

$$w_t = \psi L_{St} + \varepsilon_t \tag{38}$$

$$L_{it} = \phi^d w_t + \eta_t + u_{it} \tag{39}$$

Where w_t is the wage, $L_{St} = \sum_i S_i L_{it}$ is the size-weighted aggregate outcome (number of workers), η_t and ε_t are aggregate shocks, and u_{it} is the idiosyncratic shock to firm i.

OLS fails because aggregate wages and common shocks are correlated

We cannot estimate ψ and ϕ^d by OLS.

Why?

- ε_t and η_t are typically correlated.
- This implies that L_{St} is correlated with ε_t in the supply equation.
- And w_t is correlated with η_t in the demand equation.

GIV estimates elasticities via Two-Stage Least Squares

To estimate the causal effect of labor demand on wages (ψ), we use z_t in a standard 2SLS framework.

First Stage: Isolate exogenous variation in labor demand

$$L_{St} = \beta z_t + \text{Controls} + \nu_t$$

We regress aggregate labor demand (L_{St}) on the Granular Instrument (z_t). This extracts $\widehat{L}_{St}$, the variation driven *only* by the idiosyncratic shocks of superstar firms.

Second Stage: Estimate the causal elasticity

$$w_t = \psi \widehat{L}_{St} + \text{Controls} + \varepsilon_t$$

We regress aggregate wages (w_t) on the predicted labor demand from the First Stage ($\widehat{L}_{St}$) to obtain an unbiased estimate of ψ.

Validating the Instrument: Relevance and Exclusion

For z_t to be a valid instrument, two fundamental assumptions must hold in a granular economy.

- **Relevance (Cov$(L_{St}, z_t) \neq 0$):**

  The granular shocks must actually move the aggregate market. This requires a *concentrated market* (e.g., high Herfindahl index) where the idiosyncratic shocks of the largest firms are volatile enough to shift total labor demand.

- **Exclusion Restriction (Cov$(z_t, \varepsilon_t) = 0$):**

  The idiosyncratic shocks of large firms (z_t) must not directly affect aggregate wages (w_t) through any channel other than aggregate labor demand (L_{St}). They must be truly orthogonal to macro aggregate supply shocks (ε_t).

Constructing the Granular Instrumental Variable (z_t)

The GIV exploits the difference between size-weighted and equal-weighted averages.

Define the equal-weighted average outcome:

$$L_{Et} = \frac{1}{N} \sum_{i=1}^{N} L_{it}$$

The GIV is the size-weighted outcome minus the equal-weighted outcome:

$$z_t := L_{\Gamma t} = L_{St} - L_{Et} \tag{40}$$

The GIV isolates purely idiosyncratic firm shocks

Substitute the demand equation into the aggregate definitions:

$$\begin{aligned} L_{St} &= \phi^d w_t + \eta_t + u_{St} \\ L_{Et} &= \phi^d w_t + \eta_t + u_{Et} \end{aligned} \tag{41}$$

The common components $(\phi^d w_t + \eta_t)$ difference out:

$$z_t = u_{\Gamma t} = u_{St} - u_{Et} = \underbrace{\sum S_i L_{it}}_{\text{Size-Weighted}} - \underbrace{\frac{1}{N} \sum L_{it}}_{\text{Equal-Weighted}} \qquad \text{(Idiosyncratic Filter)}$$

z_t is made *only* of idiosyncratic shocks.

The GIV yields both supply and demand elasticities

By exogeneity, $\mathbb{E}[u_t \varepsilon_t] = 0 \implies \mathbb{E}[z_t \varepsilon_t] = 0$, where $u_t = (u_{it})_{i=1,\ldots,N}$.

Using the supply equation (38), $\mathbb{E}[(w_t - \psi L_{St}) z_t] = 0$:

$$\psi = \frac{\mathbb{E}[w_t z_t]}{\mathbb{E}[L_{St} z_t]}$$

Using the equal-weighted demand equation, $\mathbb{E}[(L_{Et} - \phi^d w_t) z_t] = 0$:

$$\phi^d = \frac{\mathbb{E}[L_{Et} z_t]}{\mathbb{E}[w_t z_t]}$$

The Full Model

Handling heterogeneous factor loadings

Superstar firms respond differently to macro trends

The simple GIV $(L_{St} - L_{Et})$ assumes every firm has the same loading $\lambda_i = 1$. If large firms are more sensitive to the Fed, the simple GIV is **"leaky"**—it correlates with η_t.

The Full Factor Model for granular firms

We now allow for heterogeneous loadings λ_i:

$$L_{it} = \phi w_t + \lambda_i \eta_t + u_{it} \qquad \text{(Full Factor Model)}$$

The simple GIV becomes:

$$z_t = (\lambda_S - \lambda_E)\eta_t + (u_{St} - u_{Et})$$

If $\lambda_S \neq \lambda_E$, the instrument is **invalid** ($\mathbb{E}[z_t \eta_t] \neq 0$).

The Solution: Orthogonal Optimal Weights Γ

We need weights Γ such that the factor components **cancel exactly**.

$$\sum_{i=1}^{N} \Gamma_i \lambda_i = 0 \quad \text{(Orthogonality Condition)}$$

This ensures $z_t = \sum \Gamma_i u_{it}$ is purely idiosyncratic.

Implementation and Assumptions

Identifying wage spillovers from superstar firms

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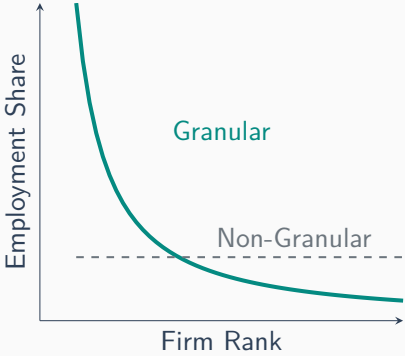
How do we apply GIVs to labor markets (e.g., tech sector)?

- **The granular shock:** Use idiosyncratic productivity or demand shocks of the largest employers.
- **The instrument:** The GIV (z_t) aggregates these large-firm idiosyncratic shocks.
- **The estimation:** Instrument local or aggregate wage changes (w_t) using the GIV to estimate the causal impact of labor demand on wages (ψ).

GIV precision relies on concentrated labor markets and volatile micro-shocks

For the instrument to be strong, we need:

1. **A large excess Herfindahl index (h):** A few superstar firms must account for a substantial fraction of total employment.
2. **Volatile idiosyncratic shocks (σ_u):** Firm-specific shocks must be large relative to aggregate macroeconomic volatility (σ_ε).



Appendix

Aggregate volatility decays at $1/\sqrt{N}$ under equal weights

Suppose there are N firms, each receiving an independent idiosyncratic shock u_{it}. Assume these shocks have mean 0 and a common variance σ^2:

$$\mathbb{E}[u_{it}] = 0, \quad \text{Var}(u_{it}) = \sigma^2, \quad \text{Cov}(u_{it}, u_{jt}) = 0 \text{ for } i \neq j$$

In a non-granular economy, the aggregate shock is an **equal-weighted** average:

$$u_{Et} = \frac{1}{N} \sum_{i=1}^{N} u_{it}$$

The variance of the aggregate shock is:

$$\text{Var}(u_{Et}) = \text{Var}\left(\frac{1}{N} \sum u_{it}\right) = \frac{1}{N^2} \sum \text{Var}(u_{it}) = \frac{N\sigma^2}{N^2} = \frac{\sigma^2}{N}$$

The aggregate volatility is therefore: $\sigma_{agg} = \frac{\sigma}{\sqrt{N}}$